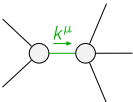
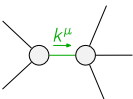


	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^-}^{\#1}\alpha\beta_X$						
$\mathcal{K}_{0^+}^{\#1}\dagger$	0	0						
$\mathcal{K}_{0^-}^{\#1}\dagger^{\alpha\beta_X}$	0	$\Upsilon_1 k^2 + 3 \binom{(2)}{K}_1$	$\mathcal{K}_{1^+}^{\#1}\alpha\beta$	$\mathcal{K}_{1^+}^{\#2}\alpha\beta$	$\mathcal{K}_{1^-}^{\#1}\alpha$	$\mathcal{K}_{1^-}^{\#2}\alpha$		
$\mathcal{K}_{1^+}^{\#1}\dagger^{\alpha\beta}$	$\Upsilon_2 k^2 + 2 \binom{(2)}{K}_1$	$\frac{-\Upsilon_3 k^2 + 4 \binom{(2)}{K}_1}{2\sqrt{2}}$	0	0				
$\mathcal{K}_{1^+}^{\#2}\dagger^{\alpha\beta}$	$\frac{-\Upsilon_3 k^2 + 4 \binom{(2)}{K}_1}{2\sqrt{2}}$	$\frac{1}{4}(-\Upsilon_4 k^2 + 4 \binom{(2)}{K}_1)$	0	0				
$\mathcal{K}_{1^-}^{\#1}\dagger^\alpha$	0	0	$\frac{\Upsilon_5 k^2}{2}$	$-\frac{\Upsilon_5 k^2}{2\sqrt{2}}$				
$\mathcal{K}_{1^-}^{\#2}\dagger^\alpha$	0	0	$-\frac{\Upsilon_5 k^2}{2\sqrt{2}}$	$\frac{\Upsilon_5 k^2}{4}$	$\mathcal{K}_{2^+}^{\#1}\alpha\beta$	$\mathcal{K}_{2^+}^{\#1}\alpha\beta_X$		
			$\mathcal{K}_{2^+}^{\#1}\dagger^{\alpha\beta}$				0	0
			$\mathcal{K}_{2^+}^{\#1}\dagger^{\alpha\beta_X}$				0	$\frac{\Upsilon_6 k^2}{2}$

	$\mathcal{T}_{0^+}^{\#1}$	$\mathcal{T}_{0^+ \alpha\beta\chi}^{\#1}$				
$\mathcal{T}_{0^+}^{\#1} \dagger$	0	0				
$\mathcal{T}_{0^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	$\frac{1}{\text{Det}(0^+)}$	$\mathcal{T}_{1^+ \alpha\beta}^{\#1}$	$\mathcal{T}_{1^+ \alpha\beta}^{\#2}$	$\mathcal{T}_{1^+ \alpha}^{\#1}$	$\mathcal{T}_{1^+ \alpha}^{\#2}$
$\mathcal{T}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{-Y_4 k^2 + 4 \frac{(2)}{\kappa_1}}{4 \text{Det}(1^+)}$	$\frac{Y_3 k^2 - 4 \frac{(2)}{\kappa_1}}{2 \sqrt{2} \text{Det}(1^+)}$	0	0		
$\mathcal{T}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$\frac{Y_3 k^2 - 4 \frac{(2)}{\kappa_1}}{2 \sqrt{2} \text{Det}(1^+)}$	$\frac{Y_2 k^2 + 2 \frac{(2)}{\kappa_1}}{\text{Det}(1^+)}$	0	0		
$\mathcal{T}_{1^+}^{\#1} \dagger^\alpha$	0	0	$\frac{2}{3 \text{Det}(1^+)}$	$-\frac{\sqrt{2}}{3 \text{Det}(1^+)}$		
$\mathcal{T}_{1^+}^{\#2} \dagger^\alpha$	0	0	$-\frac{\sqrt{2}}{3 \text{Det}(1^+)}$	$\frac{1}{3 \text{Det}(1^+)}$	$\mathcal{T}_{2^+ \alpha\beta}^{\#1}$	$\mathcal{T}_{2^+ \alpha\beta\chi}^{\#1}$
			$\mathcal{T}_{2^+}^{\#1} \dagger^{\alpha\beta}$	0	0	
			$\mathcal{T}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	$\frac{1}{\text{Det}(2^+)}$	

Abbreviations used in matrices			
$\begin{aligned} Y_1 &== \overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{16} \ \&\& Y_2 == 2 \overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_3 \ \&\& Y_3 == 2 \overset{(4)}{\kappa}_{16} - \overset{(4)}{\kappa}_4 \ \&\& Y_4 == 2 \overset{(4)}{\kappa}_{15} + 3 \overset{(4)}{\kappa}_{16} - 2 \left(\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4 \right) \ \&\& \\ Y_5 &== 2 \left(\overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_{15} \right) + \overset{(4)}{\kappa}_{16} \ \&\& Y_6 == 2 \overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16} \ \&\& \text{Det}(0^-) == k^2 \left(\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{16} \right) + 3 \overset{(2)}{\kappa}_1 \ \&\& \\ \text{Det}(1^+) &== \frac{1}{8} \left(-k^4 \left(8 \overset{(4)}{\kappa}_{15}^2 + 4 \overset{(4)}{\kappa}_{16}^2 + 4 \overset{(4)}{\kappa}_{15} \left(3 \overset{(4)}{\kappa}_{16} - 3 \overset{(4)}{\kappa}_3 - 2 \overset{(4)}{\kappa}_4 \right) + \left(2 \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4 \right)^2 - 2 \overset{(4)}{\kappa}_{16} \left(3 \overset{(4)}{\kappa}_3 + 2 \overset{(4)}{\kappa}_4 \right) \right) + \right. \\ &\quad \left. 4 k^2 \left(2 \overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16} \right) \overset{(2)}{\kappa}_1 \right) \ \&\& \text{Det}(2^-) == \frac{1}{2} k^2 \left(2 \overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16} \right) \end{aligned}$			
Lagrangian			
$\begin{aligned} &\overset{(2)}{\kappa}_1 \mathcal{K}_{\alpha\beta\chi} \mathcal{K}^{\alpha\beta\chi} - 2 \overset{(2)}{\kappa}_1 \mathcal{K}^{\alpha\beta\chi} \mathcal{K}_{\beta\alpha\chi} + \overset{(4)}{\kappa}_1 \partial_\beta \mathcal{K}^\delta_{\chi\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha\beta} - \overset{(4)}{\kappa}_1 \partial_\chi \mathcal{K}^\delta_{\beta\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha\beta} + \overset{(4)}{\kappa}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}{}^\delta + \\ &\overset{(4)}{\kappa}_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta + 2 \overset{(4)}{\kappa}_{15} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta + \overset{(4)}{\kappa}_{16} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta - \overset{(4)}{\kappa}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta - \\ &2 \overset{(4)}{\kappa}_{15} \partial^\chi \mathcal{K}^\alpha_{\alpha\beta} \partial_\delta \mathcal{K}_{\chi\beta}{}^\delta - \overset{(4)}{\kappa}_{16} \partial^\chi \mathcal{K}^\alpha_{\alpha\beta} \partial_\delta \mathcal{K}_{\chi\beta}{}^\delta - \frac{3}{2} \overset{(4)}{\kappa}_{15} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} - \frac{3}{4} \overset{(4)}{\kappa}_{16} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \\ &\frac{1}{2} \overset{(4)}{\kappa}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \frac{1}{2} \overset{(4)}{\kappa}_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \overset{(4)}{\kappa}_{15} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} + \overset{(4)}{\kappa}_{16} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} \end{aligned}$			
Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{0^+}^{\#1} == 0$	1	$\partial_\beta \mathcal{J}^\alpha_{\alpha\beta} == 0$	
$\mathcal{J}_1^{\#1\alpha} + 2 \mathcal{J}_1^{\#2\alpha} == 0$	3	$\partial_\chi \partial^\alpha \mathcal{J}^\beta_{\beta\chi}{}^\chi + \partial_\chi \partial^\chi \mathcal{J}^{\beta\alpha}_{\beta} == 3 \partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi}$	
$\mathcal{J}_2^{\#1\alpha\beta} == 0$	5	$3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \eta^{\alpha\beta} \partial_\epsilon \partial^\epsilon \partial_\delta \mathcal{J}^{\chi\chi}{}^\delta == 2 \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}^\chi_{\chi}{}^\delta + 3 \left(\partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi} \right)$	
Total # constraint(s):	9		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$-\frac{1}{2 \overset{(4)}{\kappa}_1 + 2 \overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16}}$
	1	0	$\begin{aligned} &\left(12 \overset{(4)}{\kappa}_1 \overset{(4)}{\kappa}_{15} \overset{(2)}{\kappa}_1 + 12 \overset{(4)}{\kappa}_{15}^2 \overset{(2)}{\kappa}_1 + 6 \overset{(4)}{\kappa}_1 \overset{(4)}{\kappa}_{16} \overset{(2)}{\kappa}_1 + 12 \overset{(4)}{\kappa}_{15} \overset{(4)}{\kappa}_{16} \overset{(2)}{\kappa}_1 + 3 \overset{(4)}{\kappa}_{16}^2 \overset{(2)}{\kappa}_1 + \right. \\ &160 \overset{(4)}{\kappa}_1 \overset{(4)}{\kappa}_{15}^2 p^2 + 160 \overset{(4)}{\kappa}_{15}^3 p^2 + 160 \overset{(4)}{\kappa}_1 \overset{(4)}{\kappa}_{15} \overset{(4)}{\kappa}_{16} p^2 + 240 \overset{(4)}{\kappa}_{15}^2 \overset{(4)}{\kappa}_{16} p^2 + 40 \overset{(4)}{\kappa}_1 \overset{(4)}{\kappa}_{16}^2 p^2 + \\ &120 \overset{(4)}{\kappa}_{15} \overset{(4)}{\kappa}_{16}^2 p^2 + 20 \overset{(4)}{\kappa}_{16}^3 p^2 - 216 \overset{(4)}{\kappa}_1 \overset{(4)}{\kappa}_{15} \overset{(4)}{\kappa}_3 p^2 - 216 \overset{(4)}{\kappa}_{15}^2 \overset{(4)}{\kappa}_3 p^2 - 108 \overset{(4)}{\kappa}_1 \overset{(4)}{\kappa}_{16} \overset{(4)}{\kappa}_3 p^2 - \\ &216 \overset{(4)}{\kappa}_{15} \overset{(4)}{\kappa}_{16} \overset{(4)}{\kappa}_3 p^2 - 54 \overset{(4)}{\kappa}_{16}^2 \overset{(4)}{\kappa}_3 p^2 + 72 \overset{(4)}{\kappa}_1 \overset{(4)}{\kappa}_3^2 p^2 + 72 \overset{(4)}{\kappa}_{15} \overset{(4)}{\kappa}_3^2 p^2 + 36 \overset{(4)}{\kappa}_{16} \overset{(4)}{\kappa}_3^2 p^2 - \\ &108 \overset{(4)}{\kappa}_1 \overset{(4)}{\kappa}_{15} \overset{(4)}{\kappa}_4 p^2 - 108 \overset{(4)}{\kappa}_{15}^2 \overset{(4)}{\kappa}_4 p^2 - 54 \overset{(4)}{\kappa}_1 \overset{(4)}{\kappa}_{16} \overset{(4)}{\kappa}_4 p^2 - 108 \overset{(4)}{\kappa}_{15} \overset{(4)}{\kappa}_{16} \overset{(4)}{\kappa}_4 p^2 - 27 \overset{(4)}{\kappa}_{16}^2 \overset{(4)}{\kappa}_4 p^2 + \\ &72 \overset{(4)}{\kappa}_1 \overset{(4)}{\kappa}_3 \overset{(4)}{\kappa}_4 p^2 + 72 \overset{(4)}{\kappa}_{15} \overset{(4)}{\kappa}_3 \overset{(4)}{\kappa}_4 p^2 + 36 \overset{(4)}{\kappa}_{16} \overset{(4)}{\kappa}_3 \overset{(4)}{\kappa}_4 p^2 + 18 \overset{(4)}{\kappa}_1 \overset{(4)}{\kappa}_4^2 p^2 + 18 \overset{(4)}{\kappa}_{15} \overset{(4)}{\kappa}_4^2 p^2 + \\ &9 \overset{(4)}{\kappa}_{16} \overset{(4)}{\kappa}_4^2 p^2 - \sqrt{\left(\left(2 \overset{(4)}{\kappa}_1 + 2 \overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16} \right)^2 \left(10000 \overset{(4)}{\kappa}_{15}^4 p^4 + 625 \overset{(4)}{\kappa}_{16}^4 p^4 + 81 \left(2 \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4 \right)^4 p^4 + \right.} \right. \\ &30 \overset{(4)}{\kappa}_{16}^3 p^2 \left(\overset{(2)}{\kappa}_1 - 45 \left(2 \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4 \right) p^2 \right) - 54 \overset{(4)}{\kappa}_{16} \left(2 \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4 \right)^2 p^2 \left(-\overset{(2)}{\kappa}_1 + 9 \left(2 \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4 \right) p^2 \right) + \\ &18 \overset{(4)}{\kappa}_{16}^2 \left(28 \overset{(2)}{\kappa}_1^2 - 6 \left(2 \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4 \right) \overset{(2)}{\kappa}_1 p^2 + 65 \left(2 \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4 \right)^2 p^4 \right) + \\ &80 \overset{(4)}{\kappa}_{15}^3 \left(3 \overset{(2)}{\kappa}_1 p^$